

# Cross-model uncertainty-aware minimal editing of cis-regulatory elements for cell-type-selective design

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## Abstract

**Background/Objectives:** MPRA and sequence-to-activity models have moved cis-regulatory element (CRE) research from retrospective annotation toward predictive design, and recent work has begun generating synthetic enhancers. Yet score-driven optimization can produce model-specific or heavily rewritten sequences, leaving a practical gap between prediction and compact experimental constructs. We developed SafeEdit-CRE, a cross-model, uncertainty-aware minimal-editing framework that preserves natural CRE scaffolds while producing auditable, experimentally actionable variant panels.

**Methods:** SafeEdit-CRE combines the published Malinois cell-type-specificity predictor with an architecture-diverse reviewer ensemble, validation-calibrated uncertainty, sequence-domain safeguards, and constrained beam search. Natural 200-nt CREs were edited under fixed budgets of 1, 5, 10, or 20 nucleotides; selected variants were evaluated by a separately trained multi-kernel ensemble excluded from generation, thereby separating search, review, and final evaluation.

**Results:** Across 600 held-out CRE parents in K562, HepG2, and SK-N-SH cells, cross-model specificity-margin gain increased from 0.822 for greedy editing to 0.877 for SafeEdit-CRE (paired difference 0.055; 95% CI 0.040–0.071). Against a compute-matched primary-model beam, SafeEdit-CRE maintained closely matched predicted specificity (difference  $-0.004$ ; 95% CI  $-0.011$ – $-0.004$ ) while reducing ensemble uncertainty from 0.335 to 0.314. In a locked benchmark of 90 non-overlapping parents, 60.6% of designs passed all nine prespecified sequence and model-agreement checks, compared with 38.8% for greedy and 13.1% for random substitutions (paired difference  $+21.8$  pp; 95% CI 18.3–25.3).

**Conclusions:** SafeEdit-CRE delivers a curated Tier A library of 74 reporter-screening-ready variants spanning three cell types and four edit budgets. By converting a large search space into a compact set of low-burden, cross-model supported edits, it can reduce synthesis and screening load while retaining natural sequence context. The framework therefore links regulatory-grammar analysis with a direct engineering route to synthetic regulatory elements, functional noncoding-variant prioritization, and cell-type-selective CRE design.

**Keywords:** cis-regulatory elements; transcriptional regulation; functional genomics; cell-type specificity; MPRA; regulatory-sequence design; machine learning

## 1 Introduction

Functional assays such as MPRA and increasingly capable sequence-to-activity models have moved CRE research from retrospective annotation toward predictive sequence design [1–9]. The next engineering step is not simply to find a higher-scoring sequence: it is to turn these predictions into a small number of traceable constructs that can be synthesized, compared, and iterated in the laboratory. That step remains difficult because model scores do not by themselves indicate whether an edit is robust across predictors or practical to build. CREs are particularly well suited to this bridge between biology and engineering: their transcription-factor binding sites and surrounding sequence context encode when and where genes are expressed [10–14], while their compact length permits direct synthesis and reporter measurement. A reliable design workflow could therefore support both mechanistic tests of cell-type-selective regulatory grammar and the development of tissue-selective control elements for synthetic biology, gene delivery, and functional noncoding-variant prioritization.

Current regulatory-design strategies have largely emphasized de novo generation or direct maximization of a single predictor [9, 15, 16]. They can identify high-scoring sequences, but may alter large parts of a construct or leave the natural CRE context entirely. The resulting candidates are harder to attribute to specific nucleotide changes, increase synthesis and screening burden, and provide less guidance for the next design cycle. For experimental and engineering workflows, a more useful output is a compact panel of traceable variants that preserves the parent scaffold and changes only a defined number of nucleotides.

We define this setting as *minimal editing*: increasing cell-type selectivity while retaining nearly all of the parent CRE. The parent sequence preserves native context, whereas a fixed edit budget controls design distance, construct burden, and the number of causal changes that must be interpreted. Each variant is therefore a controlled perturbation rather than an unrelated synthetic sequence. The output is explicit, auditable, and directly transferable to oligonucleotide synthesis and reporter screening, making minimal editing a practical interface between model ranking and experimental iteration.

Minimal changes alone, however, do not guarantee reliable designs. A single-predictor search can reward model-specific behavior even when the predictor performs well on held-out natural sequences [17–20]. Greedy optimization retains substitutions for their immediate gain under one objective; wider beam search explores more paths but remains vulnerable when the same model generates and ranks every candidate. The unresolved problem is therefore selection: identifying a compact set of interpretable edits whose predicted effects persist across separately trained, architecture-diverse models while uncertainty and sequence-domain behavior remain controlled. This is the point at which a predictive model becomes an engineering decision system rather than a score generator.

We developed SafeEdit-CRE to solve this selection problem (Fig. 1). The published Malinois predictor [9] supplies the primary cell-type specificity objective. During constrained beam search, an architecture-diverse residual-dilated CNN ensemble reviews proposed variants, while validation-calibrated uncertainty and safeguards for strand consistency, GC shift, homopolymers, and 6-mer naturalness filter unstable or sequence-atypical edits. A second, separately trained multi-kernel ensemble is excluded from candidate generation and used only for final cross-model evaluation. This procedural separation of search, review, and evaluation makes model agreement an explicit design criterion rather than a retrospective check. Fixed budgets of 1, 5, 10, and 20 substitutions further allow design freedom to be quantified without conflating minimal editing with unconstrained generation.

We benchmarked SafeEdit-CRE against random substitutions, iterative greedy editing, and a compute-matched primary-model beam with the same beam width and expansion schedule. Evaluation comprised cross-model generalization on 600 held-out natural CRE parents across K562, HepG2, and SK-N-SH cells; a locked nine-criterion design-quality benchmark on 90 non-overlapping parents; cell-type- and budget-stratified analyses; and complete re-search ablations. SafeEdit-CRE increased cross-model specificity margin over greedy editing, maintained closely matched predicted specificity against the compute-matched beam while lowering ensemble uncertainty, and raised the locked design-audit pass rate from 38.8% to 60.6%. The resulting Tier A library contains 74 variants across all three cell types and four edit budgets. Thus, the framework contributes both a stringent test of cross-model minimal editing and a directly usable engineering output: a compact, traceable, reporter-screening-ready panel that concentrates synthesis and experimental resources on higher-confidence CRE designs. In practical terms, it reduces the number of constructs that must be ordered and screened, while retaining enough sequence context to interpret why each candidate changes cell-type selectivity.

## 2 Materials and Methods

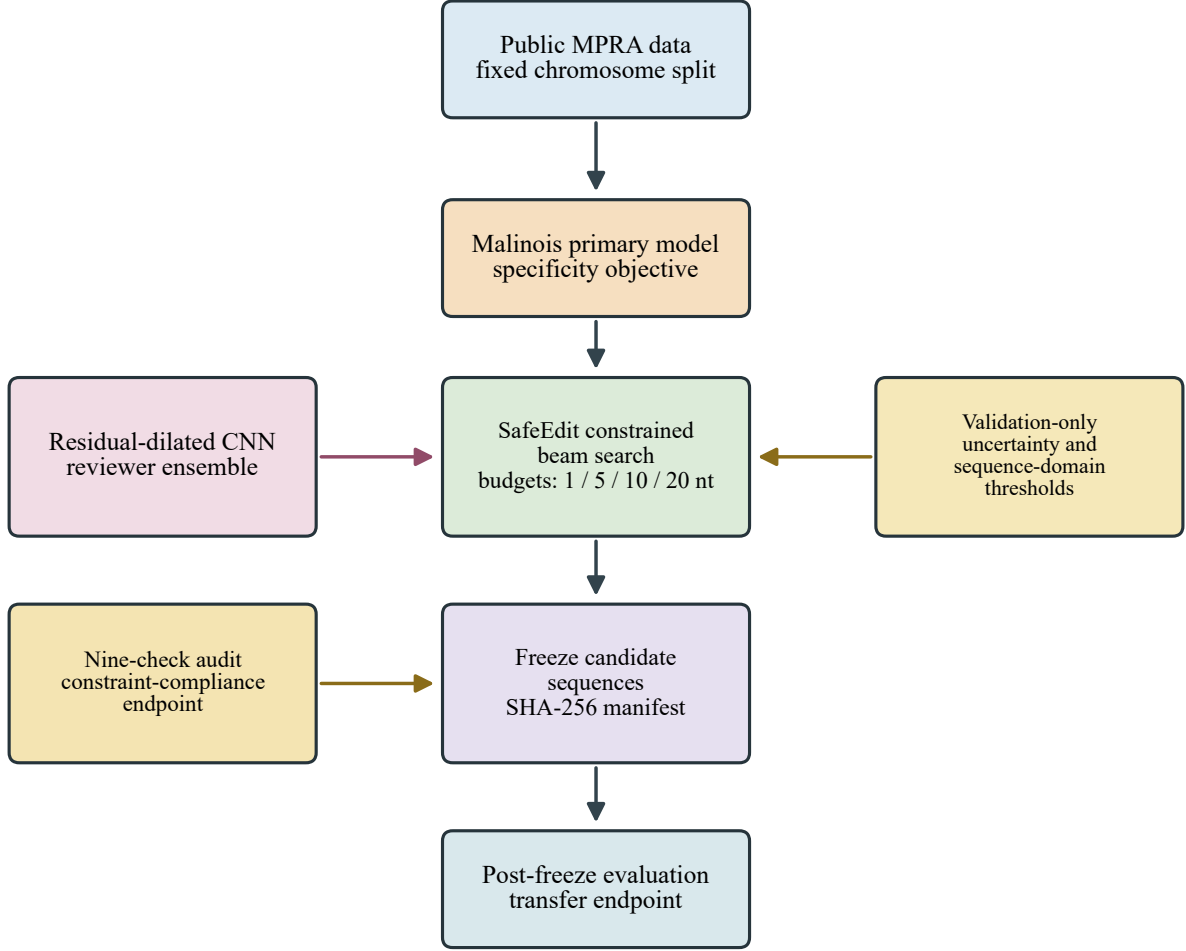
### 2.1 MPRA data and fixed chromosome partitions

We used the public MPRA sequence-performance table from Gosai et al. [9], containing activity estimates and standard errors for K562, HepG2, and SK-N-SH. The upstream chromosome partition was preserved: chromosomes 19, 21, and X were validation; chromosomes 7 and 13 were test; remaining eligible chromosomes were training. Records were required to contain only A/C/G/T, have length at most 200 nt, contain finite activity and standard-error values, and have maximum reported log-fold-change standard error below 1.0. Exact sequence duplicates were removed before full-data model training. Sequences shorter than 200 nt were centered in a 200-position tensor, and a fifth input channel marked valid positions.

### 2.2 Leakage and parent-overlap auditing

We examined exact identity, reverse-complement identity, primary-variant group membership, and cross-split Hamming distance up to three among 200-nt constructs. The near-duplicate audit identified 27 cross-split pairs involving eight training-side variant groups. Parent deny lists included earlier development and confirmation parents, matching identifiers, sequence hashes, and audit-flagged conflicts. The 600-parent primary-evaluation table was created without activity labels by sorting eligible 200-nt test sequences by SHA-256 of ID + tab + sequence and selecting the first 600 after deny-list removal. The ablation study used a fixed 180-parent subset, and the nine-criterion benchmark used a fixed 90-parent confirmatory subset after six nominal parents were found to overlap with an earlier exploratory benchmark and were excluded by identifier without reference to outcomes. Parent overlap across phases was checked by identifier and sequence hash.

The full-data loader performed exact deduplication. The separate Hamming-distance manifest was applied to parent eligibility, ensuring that no flagged near-duplicate was admitted to the editing benchmarks.



**Figure 1: SafeEdit-CRE study design.** Natural CRE parents are edited under fixed substitution budgets. Malinois supplies the primary specificity objective, while a residual-dilated CNN ensemble provides review during constrained beam search. Validation-calibrated controls for uncertainty and sequence-domain behavior guard against implausible edits. Selected sequences are then evaluated by a second separately trained multi-kernel ensemble for cross-model generalization assessment.

## 2.3 Primary predictor

Malinois BassetBranched, released with Gosai et al. [9], served as the primary predictor. The checkpoint was loaded in weights-only mode. Each sequence was evaluated in forward and reverse-complement orientations and the activity prediction was averaged. For target cell  $t$ , the specificity margin was

$$M_t(x) = f_t(x) - \max_{j \neq t} f_j(x), \quad (1)$$

where  $f_j(x)$  is predicted activity in cell type  $j$ . Candidate margin gain was  $\Delta M_t = M_t(x_{\text{edit}}) - M_t(x_{\text{parent}})$ .

## 2.4 Reviewer and cross-model evaluation ensembles

Two architecture-diverse heteroscedastic CNNs were trained on all quality-filtered training records to provide held-out evaluation at different stages. The reviewer ensemble is a residual-dilated 1D CNN with an 11-nt stem, residual blocks at dilations 1, 2, 4, and 8, and shared mean and log-variance heads; it is available during candidate search for cross-model reranking. The cross-model evaluation ensemble (termed the post-freeze evaluator in the procedural protocol below) is a multi-kernel CNN with parallel 3-, 7-, 15-, and 21-nt branches, pointwise fusion, depthwise convolution, and separate mean and variance heads [21, 22]. The use of seeded ensembles and predictive variance follows established neural-network uncertainty strategies [22–24]. Each architecture used three fixed seeds: 20260721–20260723 for the reviewer and 20260731–20260733 for the cross-model evaluation ensemble.

Models were trained for at most 20 epochs with AdamW [25], cosine learning-rate decay, batch size 256, learning rate  $10^{-3}$ , weight decay  $10^{-4}$ , dropout 0.25 [26], and gradient clipping at 5.0. The loss combined heteroscedastic

Gaussian negative log likelihood, reported measurement standard error, and a reverse-complement consistency penalty weighted 0.1. Early stopping used mean validation Pearson correlation with patience six. Test records were not used to choose checkpoints, architectures, hyperparameters, or variance scales.

For each ensemble, prediction was the mean of the three seed-specific means, in the spirit of bootstrap-aggregated ensemble prediction [27]. Predictive variance combined mean learned within-model variance with between-seed variance. One multiplicative variance scale per cell type was estimated from squared errors on validation only and then frozen [19]. The cross-model evaluation ensemble was never imported by the search code; it was applied only after candidate files and their SHA-256 hashes had been written (procedurally frozen), ensuring complete separation between generation and evaluation.

## 2.5 Editing methods

All methods were evaluated for target cells K562, HepG2, and SK-N-SH at exact substitution budgets 1, 5, 10, and 20. Positions could be edited at most once along a search path.

**Random matched edits.** Positions and non-reference alleles were sampled uniformly without replacement. The 600-parent evaluation used three independent fixed seeds (20260801–20260803). Numeric endpoints were averaged across replicates for method-level comparisons; the best random replicate was never selected.

**Greedy Malinois.** At each step, every available single-nucleotide substitution was scored by the primary specificity margin and the highest-scoring substitution was retained. No review, uncertainty, or sequence-domain constraints were enforced during search.

**Primary beam (compute-matched control).** A beam of 24 partial edit sets was expanded one substitution at a time. Up to 96 candidates per step were retained by the primary score, which was the primary margin minus 0.05 times forward/reverse strand disagreement. Beam width, maximum expansion depth, and pre-screen size exactly matched SafeEdit-CRE. No reviewer score, uncertainty penalty, or sequence-domain reranking was used.

**SafeEdit-CRE.** SafeEdit-CRE used the same primary expansion and compute budget as the primary beam. The top primary candidates were rescored by the reviewer ensemble according to

$$S(x) = M_t^{\text{primary}}(x) + 0.3M_t^{\text{reviewer}}(x) - 0.1D_{\text{strand}}(x) - 0.05U_{\text{reviewer}}(x). \quad (2)$$

Candidates were prefiltered for homopolymer length above 8 and absolute parent-relative GC change above 0.12. Reranking used validation-derived, budget-specific limits for strand disagreement, reviewer uncertainty, 6-mer naturalness change, GC change, and homopolymer length. Positive primary and reviewer margins were favored; fallback scores retained a path when no candidate passed every preferred condition. A snapshot was taken independently at each requested budget.

Sequence naturalness was the position-independent mean log likelihood of overlapping 6-mers under frequencies estimated from natural training sequences. It was used as a computational sequence-domain diagnostic during reranking.

## 2.6 Nine-criterion design benchmark

The nine-criterion benchmark (G4) used 90 test-split CRE parents with no overlap to development or primary-evaluation parents (six nominal parents were excluded after identifier-level overlap with an exploratory benchmark was discovered, without reference to outcomes). For each method–parent–target–budget combination, nine Boolean checks were recorded: non-negative primary target gain, positive primary margin gain, positive reviewer margin transfer, strand disagreement within domain, reviewer uncertainty within domain, naturalness within domain, GC change within domain, homopolymer safety, and exact edit budget. Thresholds were calibrated from validation-set random edits and locked before evaluation. Infeasible combinations were required to remain in the result table with failed status.

The primary contrast was the paired nine-criterion pass-rate difference between SafeEdit-CRE and greedy. We used a 100,000-replicate parent-cluster bootstrap for 95% confidence intervals, preserving the dependence among the 12 target–budget conditions contributed by each parent. An exact two-sided McNemar test on pooled binary outcomes was retained as a descriptive sensitivity analysis. Stratified target–budget summaries were used to examine heterogeneity.

## 2.7 Primary cross-model evaluation (600 parents)

The primary evaluation (G8) evaluated 600 label-blind, previously unused parents with random matched, greedy, primary beam, and SafeEdit-CRE methods. Search produced 43,200 sequence rows because random editing had three replicates. For method-level statistics, the three random replicates were averaged to one numeric record per parent–target–budget combination, giving 28,800 rows. This collapsed table was used only for numeric comparisons, never to select a DNA sequence. Candidate files were frozen (SHA-256 hashes written) before cross-model-ensemble inference.

The principal endpoint was cross-model specificity-margin gain. Secondary endpoints were cross-model target gain, positive-margin fraction, and cross-model ensemble uncertainty. For each contrast, metrics were first averaged over the 12 target–budget conditions within a parent; 100,000 bootstrap samples then resampled the 600 parents with replacement. Infeasible beam-search outcomes were retained as parent sequences with zero edits and zero gain.

## 2.8 Complete re-search ablation

Seven searches were independently rerun on 180 parents: primary beam, full SafeEdit-CRE, SafeEdit-CRE without reviewer score, without uncertainty penalty, without strand penalty, without naturalness reranking, and without the sequence prefilter. Unlike post-hoc filter deletion, each ablation could generate a different sequence. The cross-model evaluation ensemble was then rerun on all 15,120 candidate rows. We verified candidate keys, exact edit budgets, finite numeric values, and separate computation of cross-model target and margin gains. Paired parent-cluster bootstrap intervals used the same procedure as the primary evaluation.

## 2.9 Candidate tiering

The 3,240 nine-criterion-benchmark candidates were organized into three computational priority tiers. Tier A required SafeEdit-CRE generation, passage of all nine criteria, non-negative target gain, positive primary and reviewer margin gains, exact edit budget, no synthesis-risk flags, and location on or near the two-objective primary–reviewer Pareto frontier. Near-Pareto candidates were within 5% relative or 0.1 absolute of the frontier. Tier B passed core activity and cross-model checks but permitted Pareto suboptimality or mild naturalness deviation. Tier C failed one or more hard checks and is retained only for diagnostic analysis. Three representative Tier A examples spanning K562, HepG2, and SK-N-SH are provided in Supplementary Table S10.

## 2.10 Software, reproducibility, and reporting

Preliminary development phases were run on an NVIDIA RTX 3060 Laptop GPU. Final benchmarks were run on RTX 3080 Ti compute nodes under Slurm. Fixed seeds, environment records, code hashes, candidate hashes, checkpoint hashes, and run logs are provided in the frozen release bundle. Statistical values and figures were regenerated directly from candidate-level tables by an independent analysis script implemented with PyTorch, NumPy, pandas, and Matplotlib [28–31]. All code, environment files, minimal reproduction scripts, core statistical inputs, candidate sequence tables, and retained model checkpoints are included in the release bundle.

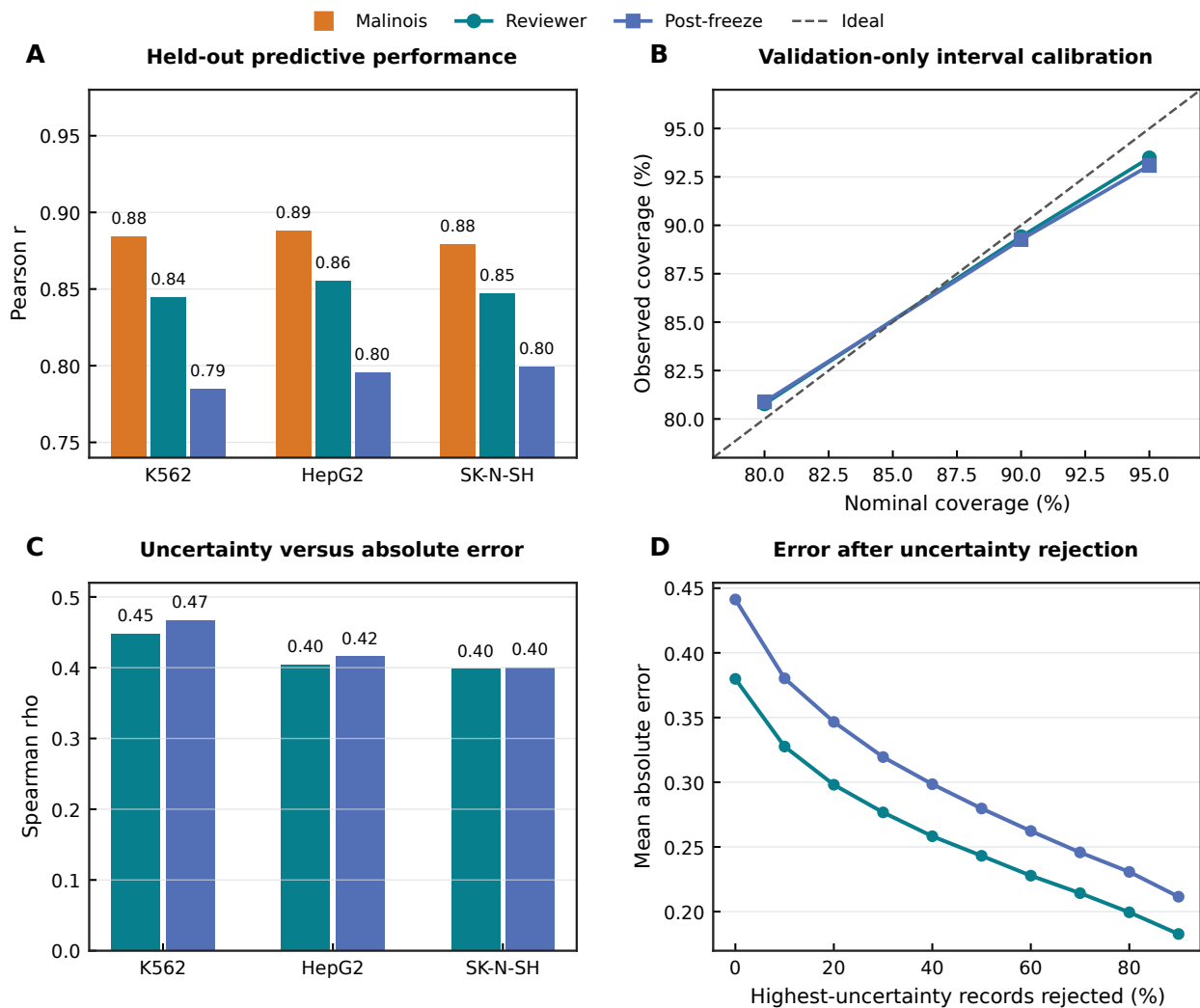
# 3 Results

## 3.1 Modeling the activity landscape for cell-type-selective CRE design

The public MPRA table contained 798,064 sequence records. After fixed quality filtering, sequence validation, length filtering, and exact-sequence deduplication, the full-data loader retained 749,054 records: 627,661 training, 58,811 validation, and 62,582 test records. Chromosomes 19, 21, and X formed validation; chromosomes 7 and 13 formed test; remaining eligible chromosomes formed training. Parent selection used identifiers and sequence hashes, never activity labels.

Forward/reverse-complement averaged Malinois predictions reproduced the published test performance (mean Pearson  $r = 0.884$  and mean Spearman  $\rho = 0.825$  across the three cell types). On the same 62,582 test records, the three-member residual-dilated reviewer ensemble achieved mean Pearson  $r = 0.849$ , whereas the three-member multi-kernel evaluator achieved  $r = 0.793$  (Fig. 2A). Their different convolutional structures allowed one ensemble to review candidates during search and the other to evaluate the frozen candidate set.

Predictive uncertainty was formed from learned heteroscedastic variance and between-seed variation, with multiplicative scaling estimated exclusively on validation data. Across cell types, 95% test-interval coverage was 0.930–0.940 for the reviewer and 0.923–0.935 for the cross-model evaluator (Fig. 2B). Reviewer uncertainty



**Figure 2: Predictive performance and uncertainty diagnostics.** **A**, test-set Pearson correlation by cell type for the primary, reviewer, and cross-model evaluator models. **B**, observed test coverage at 80%, 90%, and 95% nominal levels after validation-only variance scaling; the dashed line marks ideal calibration. **C**, Spearman association between predicted uncertainty and absolute error. **D**, mean test error after progressively rejecting the most uncertain records. All neural ensembles contain three independently seeded members.

correlated with absolute error at Spearman  $\rho = 0.399$ – $0.448$ ; cross-model-evaluator uncertainty gave  $\rho = 0.400$ – $0.467$  (Fig. 2C). Rejection curves showed decreasing mean absolute error as high-uncertainty records were removed (Fig. 2D), supporting uncertainty-guided candidate triage.

### 3.2 SafeEdit-CRE improves cross-model cell-type selectivity across 600 held-out parents

We evaluated four editing methods on 600 held-out 200-nt natural CRE parents selected by SHA-256 ordering after removal of all historical parent identifiers and sequence hashes. Methods were applied for each of three target cells (K562, HepG2, SK-N-SH) and four substitution budgets (1, 5, 10, 20 nt), producing 28,800 method-level comparisons. Random editing used three independent replicates whose metrics were averaged without best-of-three selection. After candidate generation, sequences were passed to the held-out cross-model evaluator that had no role in the search.

Across all 7,200 parent–target–budget conditions, SafeEdit-CRE achieved mean cross-model specificity-margin gain of 0.877, compared with 0.822 for iterative greedy optimization (paired difference 0.055; parent-cluster bootstrap 95% CI, 0.040–0.071) and 0.001 for random matched editing (difference 0.876; 95% CI, 0.856–0.896) (Table 1; Fig. 3A). The fraction of conditions with positive cross-model margin gain was 95.4% for SafeEdit-CRE and 97.1% for greedy, indicating that the advantage over greedy was driven by the magnitude and distribution of gains rather than by a larger fraction of positive cases. SafeEdit-CRE’s advantage therefore came from larger gains across the condition set, not from increasing the positive-case fraction.

The compute-matched primary beam search—which uses the same beam width, expansion schedule, and search



**Table 1:** Cross-model evaluation outcomes across 600 held-out parents.

| Method                         | Mean margin gain | Mean target gain | Positive margin (%) | Mean uncertainty |
|--------------------------------|------------------|------------------|---------------------|------------------|
| Random matched                 | 0.001            | 0.012            | 48.6                | 0.185            |
| Greedy Malinois                | 0.822            | 1.803            | 97.1                | 0.329            |
| Primary beam (compute-matched) | 0.881            | 1.866            | 94.8                | 0.335            |
| SafeEdit-CRE                   | 0.877            | 1.661            | 95.4                | 0.314            |

**Table 2:** Nine-criterion design benchmark across 90 non-overlapping confirmatory parents.

| Method          | Designs | Passed | Pass rate (%) | Parent-bootstrap 95% CI (%) |
|-----------------|---------|--------|---------------|-----------------------------|
| Random matched  | 1,080   | 142    | 13.1          | 10.6–15.8                   |
| Greedy Malinois | 1,080   | 419    | 38.8          | 34.6–42.9                   |
| SafeEdit-CRE    | 1,080   | 654    | 60.6          | 55.6–65.3                   |

budget as SafeEdit-CRE but relies solely on the primary Malinois predictor for candidate ranking—achieved a mean cross-model margin gain of 0.881. SafeEdit-CRE matched this performance closely (paired difference  $-0.0036$ ; 95% CI,  $-0.0114$ – $0.0043$ ; Fig. 3B) while reducing absolute target gain by 0.205 (95% CI,  $-0.229$  to  $-0.181$ ) and lowering cross-model ensemble uncertainty by 0.0209 (95% CI,  $-0.0251$  to  $-0.0166$ ) (Fig. 3C–D). Relative to greedy optimization, SafeEdit-CRE improved specificity margin by 0.055 (95% CI,  $0.040$ – $0.071$ ), reduced target gain by 0.141 (95% CI,  $-0.177$  to  $-0.106$ ), and reduced uncertainty by 0.0153 (95% CI,  $-0.0214$  to  $-0.0091$ ).

The compute-matched beam shows that search breadth alone can recover the same predicted specificity. SafeEdit-CRE changes which member of that broad candidate pool is selected: its variants have lower ensemble uncertainty and smaller target-score excursions while retaining the specificity margin.

The sequence-domain prefilter returned a zero-edit outcome for 12 of 600 parents (2.0%) in at least one condition. Retaining these outcomes in all summaries preserved the prespecified denominator and quantified the practical cost of sequence safeguards.

### 3.3 Design-criteria benchmark on 90 non-overlapping confirmatory parents

To assess whether SafeEdit-CRE designs not only score well under cross-model evaluation but also satisfy a broad set of sequence-quality and model-agreement criteria, we applied a prespecified nine-criterion design benchmark to 90 parent sequences that did not overlap with any development or primary-evaluation parents (six nominal parents were excluded after identifier-level overlap with an exploratory benchmark was discovered). For each method–parent–target–budget combination we recorded: non-negative primary target gain, positive primary margin gain, positive reviewer margin transfer, strand disagreement within domain, reviewer uncertainty within domain, naturalness within domain, GC change within domain, homopolymer safety, and exact edit budget. Thresholds were derived from validation-set random edits and locked before evaluation.

SafeEdit-CRE passed all nine criteria for 654 of 1,080 designs (60.6%), compared with 419 of 1,080 greedy designs (38.8%) and 142 of 1,080 random designs (13.1%) (Table 2; Fig. 4A). The paired SafeEdit–greedy difference was 21.8 percentage points (parent-cluster bootstrap 95% CI,  $18.3$ – $25.3$  percentage points). The SafeEdit–random difference was 47.4 percentage points (95% CI,  $43.0$ – $51.8$  percentage points). The advantage over greedy was small at a one-base budget and increased at budgets 5–20 (Fig. 4B–D).

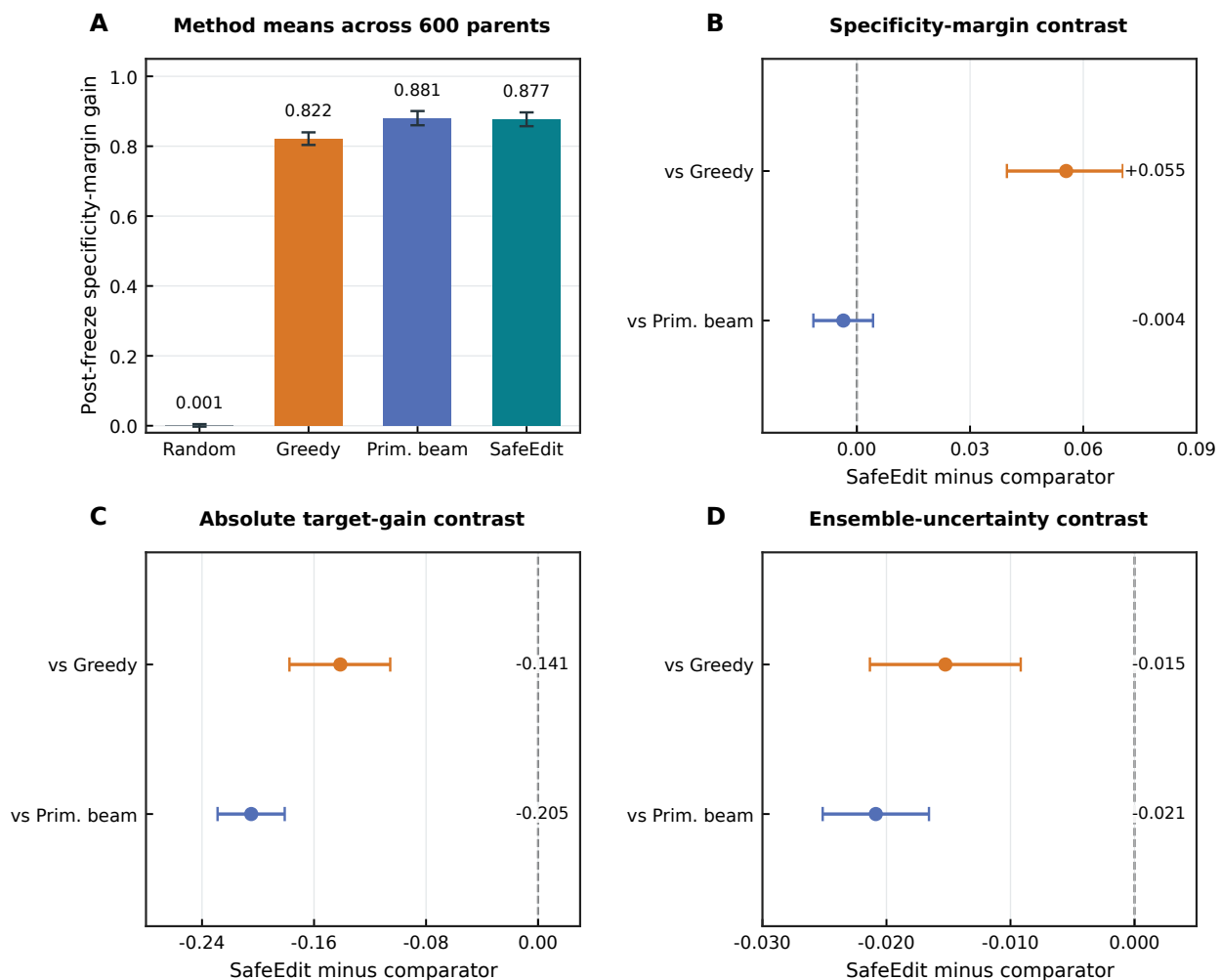
The 21.8 percentage-point difference measures the benefit of enforcing reviewer, uncertainty, and sequence-domain criteria during candidate generation. Post hoc screening of greedy designs recovered markedly fewer variants that met the same prespecified criteria.

### 3.4 Cell-type and edit-budget effects

The benefit of cross-model selection varied with target cell type and edit budget (Fig. 5). For K562, the SafeEdit–greedy cross-model margin difference progressed from  $-0.016$  at one edit to  $0.242$  at 20 edits; for HepG2, it progressed from  $-0.004$  to  $0.294$ . In SK-N-SH, differences ranged from  $-0.078$  to  $-0.033$  across budgets, indicating cell-type-specific heterogeneity in the activity landscape.

Against the compute-matched primary beam, SafeEdit-CRE differences remained within tight equivalence bounds in all strata: K562  $0.004$ – $0.031$ ; HepG2  $-0.030$ – $0.004$ ; SK-N-SH  $-0.033$ – $0.010$ , confirming that the pooled margin equivalence holds across target cells and edit budgets.

The pattern is consistent with consensus reranking becoming increasingly valuable as the number of plausible edit paths grows. At one substitution, all three directed strategies are frequently constrained to the same small



**Figure 3: Cross-model evaluation on 600 held-out natural CRE parents.** **A**, mean held-out cross-model specificity-margin gain by editing method. **B**, paired difference in margin gain for SafeEdit-CRE versus greedy and the compute-matched primary beam. **C**, corresponding paired difference in absolute target gain. **D**, corresponding paired difference in cross-model ensemble uncertainty. Points show means; intervals are 100,000-replicate parent-cluster bootstrap 95% confidence intervals. Random replicates were averaged, never selected by best outcome.

neighborhood of the parent sequence; at 5–20 substitutions, cross-model review filters out edits that look favorable under the primary predictor but transfer poorly to the held-out ensembles. The SK-N-SH result suggests that design thresholds should be calibrated per cell type rather than transferred unchanged across cellular models.

### 3.5 Complete re-search ablation identifies key components

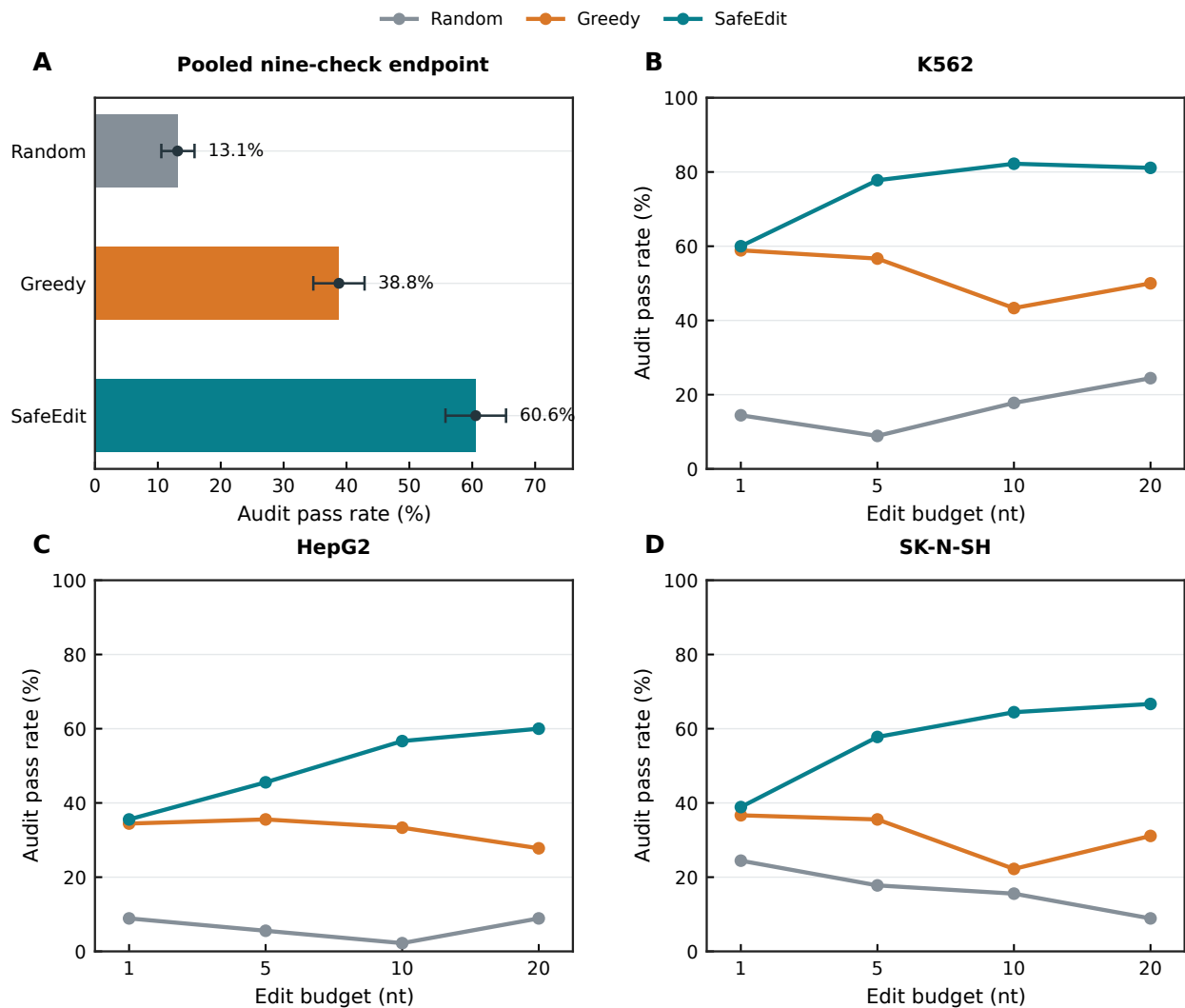
To identify which SafeEdit-CRE components drive the observed behavior, we reran the complete search under seven configurations on 180 parents (15,120 rows). Each configuration generated its own candidate sequences; filters were not removed retrospectively from a shared candidate table. Cross-model inference was recomputed for every candidate after each search.

Full SafeEdit-CRE yielded mean cross-model margin gain 0.890, target gain 1.705, and uncertainty 0.312 (Fig. 6). The matched primary beam had margin gain 0.890 (paired full-minus-primary difference 0.0002; 95% CI,  $-0.0145$ – $0.0151$ ), target gain 1.897 (difference  $-0.192$ ; 95% CI,  $-0.244$  to  $-0.143$ ), and uncertainty 0.334 (difference  $-0.0212$ ; 95% CI,  $-0.0283$  to  $-0.0140$ ).

Removing naturalness reranking raised cross-model margin gain to 0.979 and target gain to 2.017, together with an uncertainty increase to 0.343. Full SafeEdit-CRE reduced uncertainty by 0.0305 (95% CI, 0.0218–0.0394) relative to this configuration while moderating margin by 0.089 (95% CI, 0.075–0.102) and target gain by 0.312 (95% CI, 0.266–0.360). Removing the sequence-domain prefilter eliminated all 48 infeasible rows and raised margin gain to 0.910 by relaxing the GC and homopolymer limits. Strand and uncertainty penalties had smaller pooled effects, whereas the reviewer-free configuration reproduced the primary beam exactly.

Naturalness-aware reranking made the largest contribution to uncertainty reduction, whereas the sequence-domain





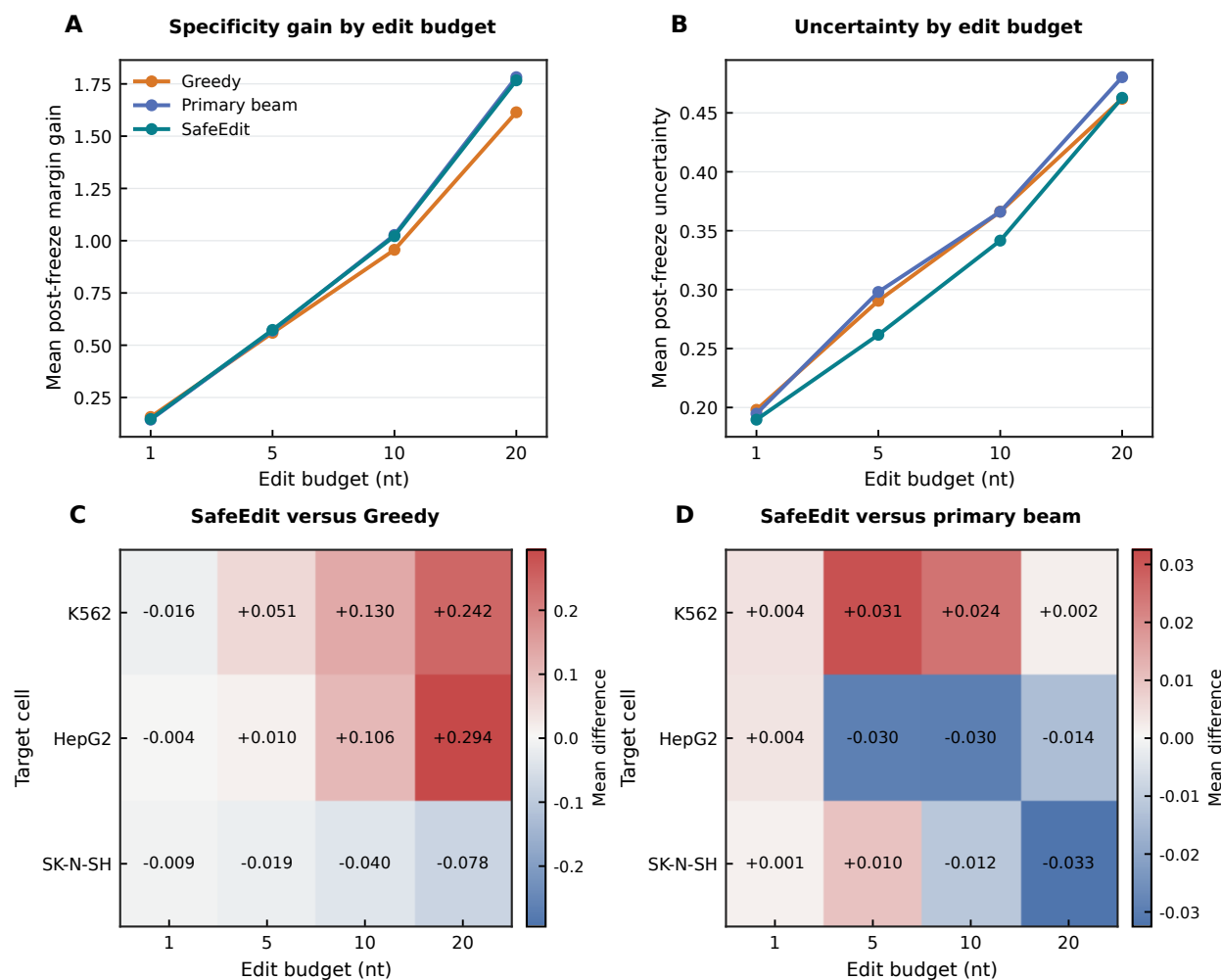
**Figure 4: Nine-criterion design benchmark.** **A**, pooled nine-criterion pass rates in the locked non-overlapping 90-parent confirmatory benchmark. Error bars are parent-cluster bootstrap 95% confidence intervals. **B–D**, pass rates stratified by edit budget for K562, HepG2, and SK-N-SH, respectively.

prefilter determined synthesis feasibility. Removing the reviewer reproduced the primary-beam result exactly, showing that the reviewer changes the returned sequence rather than merely annotating it after selection.

### 3.6 Tier A candidate library and representative examples

From the 3,240 designs in the 90-parent confirmatory benchmark, we derived a tiered candidate library for experimental screening. Seventy-four Tier A sequences were generated by SafeEdit-CRE, passed all nine criteria, met target and margin requirements, had no synthesis-risk flag, and lay on or near the primary-reviewer Pareto frontier (Supplementary Table S9; Supplementary Fig. S3). Tier B contains 1,141 alternatives that passed core activity and cross-model checks but were Pareto-dominated or mildly outside a sequence-quality criterion; the remaining 2,025 Tier C sequences are retained for failure-mode analysis. The resulting 74-sequence Tier A panel spans all three target cell types and four edit budgets, providing a compact first-pass reporter-screening set, while Tier B retains sufficient breadth for a larger secondary MPRA library.

To illustrate the nature of SafeEdit-CRE designs, Supplementary Table S10 presents three representative Tier A candidates—one each for K562, HepG2, and SK-N-SH—all generated under a 5-edit budget and achieving Pareto-frontier status without synthesis risk flags. Each example reports the parent and edited 200-nt sequence, edit positions, primary and reviewer specificity-margin gains, primary target-cell activity gain, the reduction in cross-model predictive uncertainty, GC content, naturalness, and homopolymer status. All three show positive primary target gain and positive reviewer margin transfer with decreased cross-model uncertainty. Local sequence changes near edited positions are recorded as testable sequence-grammar hypotheses for experimental follow-up.



**Figure 5: Cell-type and edit-budget effects.** **A–B**, cross-model specificity-margin gain and uncertainty pooled over the three target cells across edit budgets for greedy, primary-beam, and SafeEdit-CRE searches. **C–D**, target-by-budget differences in mean specificity-margin gain for SafeEdit-CRE relative to greedy or the compute-matched primary beam. Random editing is omitted from panels A–B because its pooled margin gain remains near zero. Lines and heatmap cells are descriptive means over parents. Edit budgets are reported in nt (nucleotide substitutions).

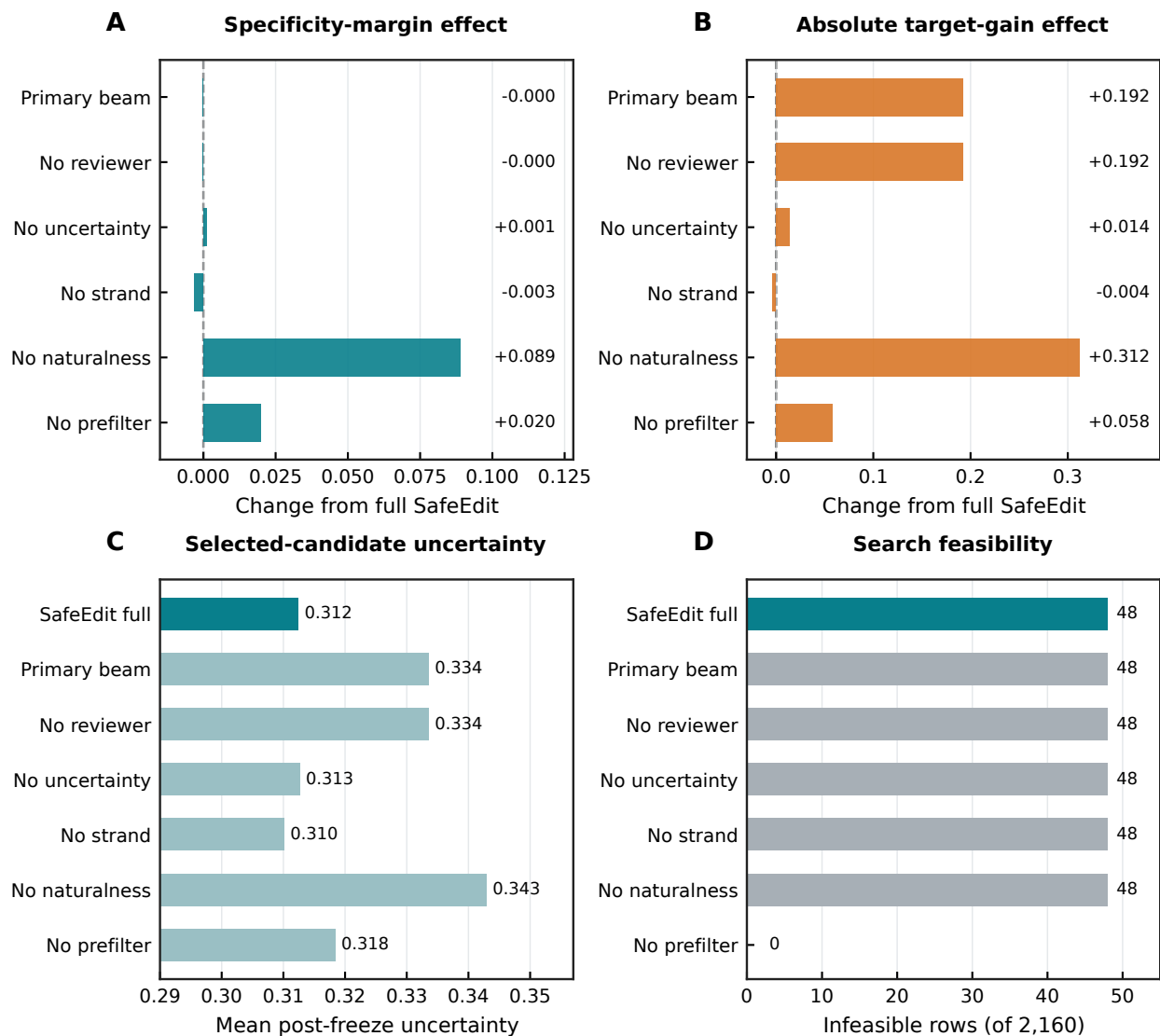
## 4 Discussion

SafeEdit-CRE was designed around a practical regulatory-genomics question: how can a natural cis-regulatory element be changed with a small number of substitutions while retaining a cell-type-selective signal when the selected sequence is inspected by a second model? Existing sequence-to-activity workflows provide strong predictors and increasingly capable design tools, but candidate ranking is often tied to the same predictor used as the optimization objective [17, 18, 20]. SafeEdit-CRE separates those roles and uses cross-model selection to decide which minimal edits are worth carrying forward.

The clearest separation from iterative greedy optimization appeared in the 600-parent evaluation. SafeEdit-CRE increased cross-model specificity-margin gain by 0.055 standardized units (95% CI 0.040–0.071), even though greedy already produced a positive margin gain in 97.1% of conditions. The difference is therefore a ranking effect among mostly plausible edits: the reviewer and uncertainty criteria favor variants with larger gains that remain consistent across models. The effect was largest at 5–20 nucleotide budgets in K562 and HepG2, where more edit paths are available and primary-model preferences can diverge.

Search breadth alone did not account for this behavior. The compute-matched primary beam achieved a margin gain of 0.881, closely matching SafeEdit-CRE's 0.877 (difference  $-0.004$ ; 95% CI  $-0.011$ – $0.004$ ). SafeEdit-CRE nevertheless returned variants with lower cross-model uncertainty (0.314 versus 0.335) and smaller target-score excursions. In other words, the framework retains the predicted cell-type-specific objective while choosing a less dispersed part of the candidate landscape. That trade-off is especially useful when synthesis and reporter capacity are limited and the experimental budget is better spent on a compact, auditable panel.

The nine-criterion benchmark gives the same result in a more demanding design setting. SafeEdit-CRE passed



**Figure 6: Complete re-search ablation on 180 parents.** **A**, change in cross-model specificity-margin gain relative to full SafeEdit-CRE. **B**, corresponding change in absolute target gain. **C**, mean cross-model ensemble uncertainty for selected candidates. **D**, number of infeasible search rows. Each configuration was independently rerun. Naturalness-aware reranking is the primary driver of uncertainty reduction, while prefilter enforcement trades margin for synthesis feasibility.

60.6% of designs, compared with 38.8% for greedy optimization, for a 21.8 percentage-point difference (95% CI 18.3–25.3). The complete re-search ablation shows why: naturalness-aware reranking makes the largest contribution to uncertainty reduction, while the sequence-domain prefilter enforces GC and homopolymer feasibility during generation. Removing the reviewer reproduced the primary-beam result exactly, identifying cross-model review as the component that changes which sequence is returned rather than merely annotating it afterwards.

The output is consequently a candidate library rather than a single model optimum. The 74 Tier A sequences passed all nine criteria, met the target and margin requirements, carried no synthesis-risk flag, and lay on or near the primary-reviewer Pareto frontier. They span three target cell types and four edit budgets, providing a compact, well-characterized first-pass panel for reporter screening and downstream regulatory-grammar dissection; 1,141 Tier B sequences retain breadth for a larger follow-up MPRA. Candidate records preserve the parent sequence, edited sequence, edit positions, model-specific gains, uncertainty, strand consistency, naturalness, base composition, homopolymer status, and edit count. The three representative examples in Supplementary Table S10 highlight concrete sequence-grammar hypotheses: localized substitutions that shift cell-type selectivity while preserving the broader CRE scaffold, directly testable via reporter assays.

Several design choices make the comparison interpretable. Parent selection was label-blind, development and evaluation parents were separated by identifiers and sequence hashes, random replicates were averaged rather than selected by outcome, and confidence intervals resampled whole parents. The post-freeze evaluator was excluded from candidate generation, while the compute-matched beam kept search width and expansion cost fixed. The

complete re-search ablation further ensures that component effects reflect different returned sequences instead of retrospective filtering of one candidate table.

The immediate biological deliverable is a compact, auditable set of CRE variants for reporter screening. The present results establish the computational ranking and its cross-model behavior; reporter assays can now test whether the predicted margin and uncertainty ordering tracks measured activity. Because the candidates retain natural CRE scaffolds and explicit edit budgets, the panel is also suitable for follow-up experiments that map local sequence changes to cell-type-selective regulatory output. Beyond reporter assays, the minimal-editing framework has direct practical implications: the Tier A panel can guide the design of cell-type-specific regulatory elements for synthetic gene circuits, help prioritize functional noncoding variants for experimental validation in disease cohorts, and provide a template for rationally modifying endogenous CREs with base or prime editing. Each candidate comes with a documented edit budget, strand-consistency checks, and naturalness scores, reducing the experimental risk of pursuing sequences that fail synthesis or produce uninterpretable results.

SafeEdit-CRE thus provides a reproducible route from sequence-to-activity prediction to experimentally testable minimal edits. Cross-model selection preserves the primary cell-type-specific objective, reduces predictive dispersion relative to a compute-matched single-model search, and produces a practical Tier A panel instead of an opaque list of unconstrained high-scoring sequences. The same selection principle can be transferred to other regulatory-prediction models and cellular contexts where candidate reliability matters as much as the nominal optimum.

## 5 Conclusions

SafeEdit-CRE establishes cross-model uncertainty-aware minimal editing as a practical route for converting sequence-to-activity models into compact, cell-type-selective CRE variant libraries. By retaining natural sequence scaffolds and explicitly controlling edit distance, model agreement, and predictive uncertainty, the framework provides a focused starting point for functional-genomics studies of regulatory sequence grammar.

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## Institutional Review Board Statement

Not applicable. The study used publicly available sequence and MPRA summary data and did not involve human participants, identifiable personal data, or animal experiments.

## Informed Consent Statement

Not applicable.

## Data Availability Statement

The processed MPRA measurements used to train Malinois are provided in Supplementary Table 2 of Gosai et al. [9]. The associated raw data, processing notebooks, model weights, and immunofluorescence images are archived at Zenodo record 10698014. All SafeEdit-CRE code, environment files, frozen configuration manifests, parent identifiers, candidate-level result tables, statistical summaries, analysis scripts, fixed model seeds, and the 74 Tier A candidate sequences are assembled in the upload-ready v1.1.0 release bundle. The bundle will be deposited in a permanent Zenodo record and mirrored at <https://github.com/Haangyeon/SafeEdit-CRE> before publication; the final DOI and commit hash will be inserted in the published version. During peer review, the complete release bundle is available from the corresponding author. It includes a minimal end-to-end reproduction script that regenerates the main figures and statistical tables from the released summary inputs.

## Author contributions

Angran Xia conceived the study, performed the computational analyses, prepared the figures and manuscript, and approved the submitted version.

## Competing interests

The authors declare no competing interests.

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## Generative AI Use Statement

Generative AI tools were used for language editing, document organization, and manuscript-formatting assistance. No generative AI system was used to generate the experimental or computational data, perform the statistical analyses, or replace the author's scientific judgment.

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